

## H: HCubes Costs

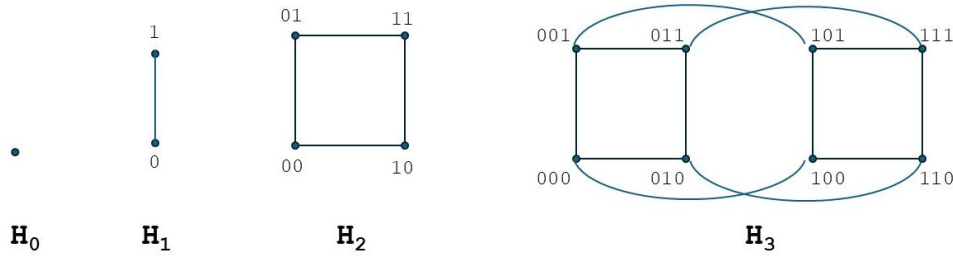
Source file name: `hcubes.c`, `hcubes.cpp`, `hcubes.java`, or `hcubes.py`

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An  $n$ -hypercube,  $n \geq 0$ , is a data structure with numerous applications in databases and communications. It is defined recursively as follows<sup>2</sup>:

- The 0-hypercube  $H_0$  is the undirected graph with one vertex denoted by the empty string  $\langle \rangle$  and 0 edges.
- For  $n \geq 0$ , the  $(n + 1)$ -hypercube  $H_{n+1}$  is the graph consisting of two copies of  $H_n$ , with vertex sets  $V_{n+1} = 0V_n \cup 1V_n$  and edges in each of the copies  $H_n$ , plus one edge  $(0v, 1v)$  for each  $v \in H_n$ .

It is rather easy to see that  $V_n$  is exactly the set of all the binary strings with length  $n$ , for  $n \geq 0$ . The figure illustrates the representation of  $H_n$ , for  $n \leq 3$ :



InterConnectivity Per Cubes (ICPC) is a start-up that uses  $n$ -hypercubes in network designs. Nowadays they are solving core software to determine minimum costs for data communication between pairs of nodes of an  $n$ -hypercube network. More precisely, the cost of a communication between nodes  $a, b \in H_n$  is the cost of a minimum cost path between the nodes. The cost of a path is the sum of the costs of its edges. For  $u, v \in V_n$ , so that there is an edge  $(u, v) \in H_n$ , let  $c_n(u, v)$  the cost of this edge. ICPC models defines recursively the costs of the edges so that:

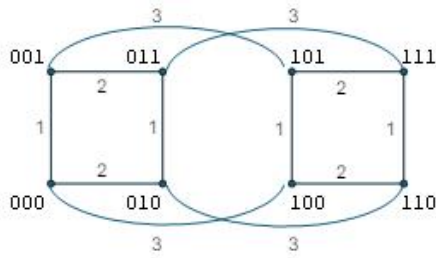
- $c_1(0, 1) = 1$
- $c_n(u, v) = c_n(v, u)$  for  $u, v \in V_n, n > 0$
- $c_{n+1}(xu, xv) = c_n(u, v)$ , if  $x \in \{0, 1\}, u, v \in V_n, n > 0$
- $c_{n+1}(xu, yu) = n + 1$ , if  $x, y \in \{0, 1\}, u \in V_n, x \neq y, n > 0$

For instance,  $H_3$  has the following arc costs:

Since every node has a unique  $n$ -bit string that denotes it, each node  $u$  can be denoted with the integer value that represents, for instance, in decimal notation. For example, in  $H_3$  the nodes 110 and 010 can be denoted with the integers 6 and 2, respectively.

Your task is to help ICPC with the problem of determining the cost of a minimum cost path of a communication between two nodes of their hypercubes. For instance, in the case of the above described  $H_3$ , the cost of a communication between nodes 6 and 2 is 3, between 1 and 7 is 5, etc.

<sup>2</sup>Binary strings are used to denote nodes of hypercubes. Some notation about binary strings: for  $x \in \{0, 1\}$  and  $\alpha$  a binary string,  $x\alpha$  is the binary string adding the bit  $x$  before  $\alpha$ . If  $A$  is a set of binary strings and  $x \in \{0, 1\}$ , then  $xA$  denotes the set of binary strings of the form  $x\alpha$ , where  $\alpha \in A$ .



### Input

The input consists of several test cases. A case begins with a line with three integer numbers  $n$ ,  $a$  and  $b$ ,  $0 < n \leq 20$ ,  $0 \leq a, b < 2^n$ , where  $a$  and  $b$  are decimal denotations of vertexes of the corresponding  $n$ -hypercube.

The input end is signaled with a line (that should not be processed)

0 0 0

*The input must be read from standard input.*

### Output

For each test case, print one answer line with an integer number corresponding to the cost of a minimum cost path from  $a$  to  $b$  in the  $n$ -hypercube.

*The output must be written to standard output.*

| Sample Input | Sample Output |
|--------------|---------------|
| 3 6 2        | 3             |
| 3 1 6        | 6             |
| 2 3 3        | 0             |
| 1 0 1        | 1             |
| 0 0 0        |               |